EvolvingBrain: An Adaptive Skill System for Multi-modal Agents Self-Evolution in Open Worlds

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HarnessRSI 68 ★★

evidence: retained

Paper

1 What changes, what is kept

1 Collect trajectories
own runs per §2.2, but released ones on VAB (A.1); successes and failures kept

2 Extract two pattern types
procedural = what advances the task;
problematic = what blocks it (images checked)

3 Trap Skills + retry
retry same task from its start with the skill; revise till success or budget is out

≤ 3 iters

4 Accept only clean runs
success, no recurring or new problem, else back to step 2

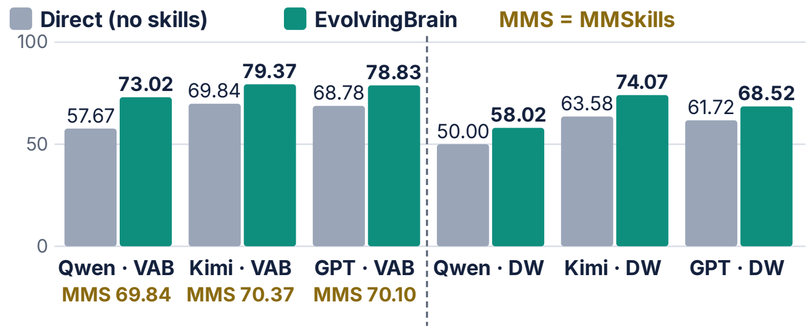
5 Distill skills + graph
accepted runs → Plan + Action Skills, then Domain Skills; link the graph

Test time: library frozen, weights never change

Memory	State gate	Fuse
progress, blockers	none / reuse / retrieve	temporary guidance

guidance is never written back to the library

2 Headline: success rate vs direct agent



Test splits, mean of three runs (Table 2), no SD or CI. VAB = VAB-Minecraft; DW = DiscoveryWorld in-domain test (MMSkills not run there).

3 Controls that decide if the gain is real

Ablations, same backbone (Tables 2, 4, 6)

Success %	Direct	1 iter	no Trap	full
Qwen · VAB	57.67	68.68	70.90	**73.02**
Qwen · DW	50.00	54.63	55.02	**58.02**
Kimi · VAB	69.84	75.29	75.27	**79.37**
Kimi · DW	63.58	69.44	69.77	**74.07**
GPT · VAB	68.78	75.57	74.58	**78.83**
GPT · DW	61.72	65.74	65.72	**68.52**

1 iter = one-pass build; no Trap = Trap retrieval off. No SD.

Retrieval depth, DW = DiscoveryWorld (Table 5)

Success %	coarse	fine	adaptive
Qwen · VAB	66.38	72.06	**73.02**
Qwen · DW	53.70	57.41	**58.02**
Kimi · VAB	78.21	74.18	**79.37**
Kimi · DW	72.84	68.52	**74.07**

Only Qwen and Kimi tested; fine detail helps Qwen, hurts Kimi.

Held-out themes, success % (Table 7)

2 unseen themes	Direct	full	gain pp
Qwen3.8-27B	36.67	46.67	**10.00**
Kimi-K3	48.89	60.00	**11.11**
GPT-5.6-Sol	46.67	56.67	**10.00**

In-to-held-out drop 12.42 vs 14.36 pp: gap 1.94‡ < one task per run (3.33‡). Gains in-domain, DW: 8.02‡, 10.49‡, 6.80‡; VAB: 15.35‡, 9.53, 10.05.

Qwen, Kimi: xhigh setting; GPT: high. ‡ = computed by us (SOURCE.md).

4 Where the evidence is thin

- **One skill baseline:** MMSkills (released skills, VAB only). Qwen 73.02 vs 69.84; Kimi, GPT: MMSkills ≈ Direct
- **No variance;** "three runs" undefined. Adaptive beats the best fixed depth by 0.61‡–1.23‡ pp
- **Small sets:** held-out = 2 themes, 30 tasks (10.00 pp ≈ 3 tasks‡); test sizes do not match Table 2
- **No cost reported** for building the library or for extra LLM calls per round; Direct gets only the same round budget
- Not ablated: memory, gate, no-Domain variant. Code not public

Takeaway

Skills distilled from a frozen model's own successes and failures, then fetched by execution state, beat the same model without skills on every backbone and both benchmarks, and the library works on held-out themes. The evidence is single averaged numbers: no variance, no cost, one skill baseline.

Our read · HarnessRSI

Score 68 · ★★ · evidence level: retained

Reports: held-out themes, ablations, same round budget.
Does not report: variance, cost, matched-compute baselines. Code is announced, not public.